



United International University
Summer 2026, Introduction to Computer Systems: CSE 1110
Problem Set (Beginner to Advanced)
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Beginner Level (Warmup)

1. Write a C program that accepts two integers from the user and calculates the product of the two integers.
2. Write a C program that accepts an employee's ID, total worked hours in a month and the amount he received per hour. Print the ID and salary
3. Write a C program to calculate the distance between two points.
4. Write a C program to find if an integer is even or odd.
5. Write a C program to check if an integer is positive, negative or zero.
6. Write a C program where the user will give his marks (ranging from 0 to 100.0) as input and you will calculate his grade based on your university grading system and show it as output.
7. Write a C program to check if a character is alphabet, numeric or special character.
8. Write a C program that reads three floating-point values and checks if it is possible to make a triangle with them. Determine the perimeter of the triangle if the given values are valid.
9. Write a C program that reads an integer between 1 and 12 and prints the month of the year in English.
10. Write a C program to accept a coordinate point in an XY coordinate system and determine in which quadrant the coordinate point lies.

Intermediate Level

11. Write a C program to find the largest of three numbers.
12. Write a C program that reads 5 numbers and counts the number of positive numbers and negative numbers.
13. Write a C program that accepts some integers from the user and finds the highest value and the input position.
14. Write a C program to find whether a given year is a leap year or not.
15. Write a C program to calculate the root of a quadratic equation.
16. Write a C program to read temperature in centigrade and display a suitable message according to the temperature state below:
Temp < 0 then Freezing weather
Temp 0-10 then Very Cold weather
Temp 10-20 then Cold weather
Temp 20-30 then Normal in Temp
Temp 30-40 then Its Hot
Temp >=40 then Its Very Hot
17. Write a C program to check whether a triangle is Equilateral, Isosceles or Scalene.

18. Write a program in C to calculate and print the electricity bill of a given customer. The customer ID, name, and unit consumed by the user should be captured from the keyboard to display the total amount to be paid to the customer.

The charge are as follow :

Unit	Charge/unit
upto 199	@1.20
200 and above but less than 400	@1.50
400 and above but less than 600	@1.80
600 and above	@2.00

19. Write a C program to take input n numbers from the user and find the sum of those n numbers.
20. Write a C program to display the n terms of odd natural numbers and their sum.
21. Write a C program to calculate the factorial of a given number.
22. Write a program in C to display the n terms of a harmonic series and their sum.
23. $1 + 1/2 + 1/3 + 1/4 + 1/5 \dots 1/n$ terms
24. Write a program in C to find the sum of the series $[x - x^3 + x^5 + \dots]$.
25. Write a C program that displays the n terms of square natural numbers and their sum.
26. Swap the values of two variables using a third variable.
27. Swap the values of two variables without using a third variable.
28. Find the maximum and minimum of n numbers given by the user.
29. Read a positive integer and print the sum of its digits.
30. Read an integer and decide whether it is prime or not.
31. Write a C program to find the factors of a given number.
32. Write a program in C to display a given number in reverse order.
33. Write a C program to check whether a number is a palindrome or not.

Semi-Advanced Level

1. Write a program in C to find the number and sum of all integers between 100 and 200 which are divisible by 9.
2. Write a C program to check whether a given number is a 'Perfect' number or not.
3. Write a C program to check whether a given number is an Armstrong number or not.
4. Write a program in C to display the first n terms of the Fibonacci series.
5. Write a C program to find the second largest number from n input numbers.
6. Write a C program where users give n-1 natural numbers ranging from 1 to n as input where the user does not repeat a number. You need to find out the missing number within the range 1 to n.

7. You are given three digits a, b, c. Two of them are equal, but the third one is different from the other two.
8. Find the Greatest Common Divisor (GCD) of two numbers.
9. Find the Lowest Common Multiple (LCM) of two numbers.

Advanced Level (For Starting Your Journey as a Contest Programmer)

1. Write a program in C to check whether a number can be expressed as the sum of two primes.
2. Write a C program to check whether a given number is ugly or not. An ugly number is a positive integer whose only prime factors are 2, 3, and 5.
3. You are given two integers a and b. In one move, you can choose some integer k from 1 to 10 and add it to a or subtract it from a. In other words, you choose an integer $k \in [1; 10]$ and perform $a := a + k$ or $a := a - k$. You may use different values of k in different moves. Your task is to find the minimum number of moves required to obtain b from a.
4. In this problem you are to calculate the sum of all integers from 1 to n, but you should take all powers of two with minus in the sum. For example, for $n = 4$ the sum is equal to $-1 - 2 + 3 - 4 = -4$, because 1, 2 and 4 are 2^0 , 2^1 and 2^2 respectively.
5. Polycarpus adores TV series. Right now he is ready to finish watching a season of a popular sitcom "Graph Theory". In total, the season has n episodes, numbered with integers from 1 to n. Polycarpus watches episodes not one by one but in a random order. He has already watched all the episodes except for one. Which episode has Polycarpus forgotten to watch?

Examples

input	Copy
10 3 8 10 1 7 9 6 5 2	
output	Copy
4	

6. <https://codeforces.com/contest/387/problem/A>
7. <https://codeforces.com/contest/621/problem/A>
8. <https://codeforces.com/contest/606/problem/A>
9. <https://codeforces.com/contest/570/problem/A>
10. <https://codeforces.com/contest/534/problem/A>